Started on	Wednesday, 30 September 2026, 9:11 PM
State	Finished
Completed on	Wednesday, 30 September 2026, 9:23 PM
Time taken	12 mins 16 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as “No”.

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value “x”)

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int findPair(int arr[],int left,int right,int x,int*a,int*b)
{
    if(left >= right)
        return 0;

    if(arr[left] + arr[right] == x)
    {
        *a = arr[left];
        *b = arr[right];
        return 1;

    }

    if(arr[left] + arr[right] < x)
        return findPair(arr,left + 1,right,x,a,b);
```

```
    else
        return findPair(arr,left,right - 1,x,a,b);
}

int main()
{
    int n,x,i;
    int arr[100];
    int a,b;
    scanf("%d",&n);

    for(i= 0;i<n;i++)
        scanf("%d",&arr[i]);

    scanf("%d",&x);

    if(findPair(arr,0,n-1,x,&a,&b))
    {
        printf("%d\n",a);
        printf("%d\n",b);
    }
    else
    {
        printf("No\n");
    }
    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.